Praveen Bajpai

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Education

- **IIIT Naya Raipur** — Naya Raipur, Chhattisgarh
  B.Tech in Electronics & Communication Engineering — *Aug 2024 – May 2028*
  CGPA: **8.82 / 10.0**
- **Lucknow Public School** — Lucknow, Uttar Pradesh
  Class XII: ***90%*** | *Class X:* ***95%*** — *2023 | 2021*

Experience

- **Samsung R&D Institute India** — India
  AI Research Intern – Agentic AI & LLM Systems — *Dec 2025 – Jan 2026*
  Built an end-to-end **Agentic AI pipeline** (Claude Agent SDK + FastAPI) converting Figma designs into production-ready React + TypeScript components – **92% faster design-to-code** with **100% build success**.
  Integrated **LLM-based agents** with the frontend via REST APIs for intelligent task automation and decision-making; worked hands-on with **Agentic AI, MCP servers and RAG pipelines** in production.
  Performed model evaluation and experimentation, cutting monthly operating cost by ∼**90%** by replacing local GPU inference with cloud LLMs via a LiteLLM proxy.
- **Rootvestors** — Remote
  Backend Developer Intern — *Jun 2025 – Nov 2025*
  Built the backend of an **adaptive learning platform** for NCERT Chemistry (Classes 11 & 12) using **Node.js, TypeScript and Express.js**, serving AI-personalized lessons via the **Anthropic Claude API**.
  Designed a dual-database architecture – **PostgreSQL (Neon) + Prisma ORM** for students and quizzes, plus a **Neo4j knowledge graph** of 28 chapters and 140+ subtopics modeling prerequisites.
  Implemented **EMA-based mastery tracking**, quiz scoring, subtopic unlock logic (0.6 mastery gate), Zod validation and per-student rate limiting; exposed **20+ REST endpoints** for content, quizzes and recommendations.

Projects

- **The CinePrism – Film Criticism Journal** | *Next.js, React, Tailwind CSS, Vercel* — thecineprism.com
  Built a full-stack editorial journal for film reviews, essays and curated lists, grown to **10,000+ monthly visitors**.
  Implemented end-to-end SEO – canonical URLs, OpenGraph/Twitter cards and semantic markup – with a custom long-form, typography-first reading experience.
  Shipped fast, fully responsive pages on **Next.js + Vercel** with statically generated content and optimized asset loading.
- **AI-Powered Image & Video Editor** | *Next.js, Python, TensorFlow.js, FFmpeg, AWS Lambda*
  Built an AI media editor with **background removal, object and face detection** via in-browser **TensorFlow.js** inference.
  Added real-time filter previews and color grading, plus video tools for **trimming, merging and quality-configurable exports** powered by FFmpeg.
  Offloaded heavy processing to **Python + AWS Lambda** serverless jobs, keeping the Next.js UI responsive during exports.

Technical Skills

- **Languages:** C++, Python, JavaScript, TypeScript, C, SQL
- **Core:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, REST APIs
- **Frameworks:** React.js, Next.js, Node.js, Express.js, Flask, FastAPI
- **AI & Data:** Anthropic Claude API, Agentic AI, RAG, TensorFlow.js, Prisma, PostgreSQL, Neo4j
- **Tools:** Git, FFmpeg, AWS Lambda, Vercel, VS Code

Achievements

- Secured **98 percentile in JEE Mains** among over 1.2 million candidates.
- Secured **CRL 14,000 in JEE Advanced**, among India's top engineering entrance candidates.
- Solved **750+ problems** across LeetCode, Codeforces and GeeksforGeeks covering DP, graphs, trees, greedy and backtracking.
- Selected for **ATF 2025**, demonstrating strong analytical and problem-solving skills.